KHANH DUY TRAN

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Google Scholar | ORCID | Portfolio

Research Interests

My research focuses on the technical feasibility of edge-enabled AI for wearable biomedical systems that support cardiovascular monitoring outside clinical settings. I specifically work on **(1) wearable biomedical systems engineering**, spanning hardware design, physiological signal processing, embedded AI deployment, and cloud/system integration; and **(2) reliability-aware clinical AI**, including predictive uncertainty, selective prediction, risk–coverage analysis, and identification of low-confidence diagnostic outputs.

Education

M.Eng. in Electrical Engineering, International University – Vietnam National University, HCMC *2024–Present*
Expected degree completion: November 2026–March 2027 | *English-medium instruction*
Master's Thesis: *Edge AI-Enabled Smart Healthcare Framework for Real-Time Cardiovascular Disease Detection*

B.Eng. in Electrical Engineering, International University – Vietnam National University, HCMC *2020–2024*
GPA: 3.75/4.0 — (ranked 1st in class K20) | *English-medium instruction* | *ABET-accredited*
Undergraduate Thesis: *Development of a Real-Time Automatic Cardiovascular Disease Detection Edge Computing System*

Research Experience

Research Assistant, School of Electrical Engineering, International University – VNUHCM *2022–Present*
Advisor: Dr. Quoc T. Huynh

- Conducted research on wearable biomedical systems spanning hardware fabrication, physiological signal processing, and embedded AI, with applications involving electrocardiography (ECG), electroencephalography (EEG), seismocardiography (SCG), and phonocardiography (PCG).
- Developed end-to-end wearable monitoring platforms demonstrating **system-level viability** through multi-lead ECG sensing, real-time signal processing, on-device inference, local data management, wireless communication, and cloud-based monitoring.
- Developed and evaluated new signal-processing and AI methods for clinically relevant tasks, including arrhythmia classification, cardiovascular and neurological abnormality detection, heart-sound analysis, and SCG fiducial-point identification, with emphasis on diagnostic reliability and deployment constraints.
- Supported collaborative research through experimental design, implementation, technical review, research mentoring, report preparation, and manuscript writing and revision.

Publications

Journal Articles and Letters (Under Review)

[1] **Tran, K.D.**, Tong, H.D., Doan, M.N., Vu, L.T.K., Nguyen, U.D., and Huynh, Q.T. "An AGENet-Based Cloud-Enabled Wearable Telemedicine System for AI-Assisted ECG Monitoring and Arrhythmia Classification." *IEEE Open Journal of Engineering in Medicine and Biology*, submitted May 2026; under review.

[2] **Tran, K.D.** "Reliability of Predictive Uncertainty in Free-Living Wearable ECG Signal-Quality Assessment under Noise and Distribution Shift." *Medical Engineering & Physics*, submitted Aug. 2026; under review.
Preprint: *medRxiv*, doi:10.64898/2026.08.25.26361304.

[3] **Tran, K.D.** "Sensing-Configuration Sufficiency and Source-Specific Information in Wearable Human Activity Recognition." *IEEE Sensors Letters*, submitted Sep. 2026; under review.

Conference Papers

[1] **Tran, K.D.**, Tang, L.H., Nguyen, U.D., and Huynh, Q.T. "Developing a Deep Learning Model Using a Combination of CNN and RCNN for Cardiovascular Disease Classification." *ATC 2024*, Ho Chi Minh City, Vietnam, Oct. 2024. doi:10.1109/ATC63255.2024.10908251.

[2] **Tran, K.D.**, Tang, L.H., Nguyen, U.D., and Huynh, Q.T. "Developing a Mixed Deep Learning Model Approach for Arrhythmia Classification Based on ECG Signal." *RIVF 2024*, Da Nang, Vietnam, Oct. 2024. doi:10.1109/RIVF64335.2024.11009042.

[3] Phan, T.L.N., **Tran, K.D.**, Doan, M.N., Nguyen, U.D., Vu, L.T.K., and Huynh, Q.T. "Robust Abnormal Heart Sound Detection via Soft Voting Ensemble on Augmented MFCC Representations." *RIVF 2025*, Ho Chi Minh City, Vietnam, Nov. 2025. doi:10.1109/RIVF68649.2025.11365204.

[4] Doan, M.N., **Tran, K.D.**, Phan, T.L.N., Son, H.D., Nguyen, U.D., Vu, L.T.K., and Huynh, Q.T. "EEG-Based Brain–Computer Interface for Directional Control Using Machine Learning on Overt Spatial Attention." *RIVF 2025*, Ho Chi Minh City, Vietnam, Nov. 2025. doi:10.1109/RIVF68649.2025.11365260.

Grants and Funding

- **Researcher**, "Cloud-Enabled Autonomous Telemedicine Platform for Continuous ECG Monitoring and Arrhythmia Screening," faculty research grant, School of Electrical Engineering, Vietnam National University, Ho Chi Minh City (VNU-HCM). Grant No. T2026-01-EE, 2026. (PI: Dr. Quoc T. Huynh)
- **Researcher**, "Developing Deep Learning Models for Arrhythmia Classification," faculty research grant, School of Electrical Engineering, Vietnam National University, Ho Chi Minh City (VNU-HCM). Grant No. T2023-03-EE, 2024. (PI: Dr. Quoc T. Huynh)
- **Student Principal Investigator**, "Developing a Cardiovascular Diseases Monitoring System," student research grant, International University – VNUHCM, 2024.

Honors and Awards

- Class of K20 Valedictorian, School of Electrical Engineering, International University (2024).
- Second Prize, Fifth Scientific Conference for Students, International University – VNUHCM (2024).
- Euréka Research Awards (2023, 2025); Bosch Talented Scholarship Award (2023); Viet Nam Electricity Scholarship Award (2024).
- University Scholarship recipient (2020–Present).

Teaching & Industry Experience

Teaching Assistant, International University – VNUHCM *2023–2024*

- Supported C/C++ Programming, Object-Oriented Programming, and Introduction to Computing.

Associate Software Engineer, OPSWAT *2024–2025*

- Developed C++/Linux applications for malware detection and system scanning using multithreading, multi-process execution, process isolation, logging, and network analysis.

Software Engineer Intern, DEK Technologies *2023*

- Developed microservice-based messaging features using WebSockets, Linux IPC/APIs, and Docker.

Peer Review

Reviewer, *Biomedical Physics & Engineering Express* (IOP Publishing), 2026.

Technical Skills

Programming	Python, C/C++, MATLAB, Bash, HTML/CSS, JavaScript
ML/DL	PyTorch, TensorFlow, Keras, scikit-learn
Embedded & Hardware	STM32, ESP32, Raspberry Pi
Tools & Systems	Linux, Windows, Git, Docker, Altium, LaTeX
Languages	English, Vietnamese